

Name:

Maths Class:

Year 11

Mathematics Extension 1

Course

Assessment 3

September, 2022

Time allowed: 90 minutes

General Instructions:

- Marks for each question are indicated on the question.
- Approved calculators may be used
- All necessary working should be shown
- Full marks may not be awarded for careless work or illegible writing
- ***Begin each question on a new page***
- Write using black or blue pen
- All answers are to be in the writing booklet provided
- A NESA reference sheet is provided.

Section 1 Multiple Choice

Questions 1-6

6 Marks

Allow approximately 10 minutes for this section

Section II Questions 7 – 12

60 Marks

Section 1

6 Marks

Allow about 10 minutes for this section.

Attempt all questions

Answer each question on the multiple-choice page in your answer booklet.

Question 1

Which of the following is the solution to $3 \leq |x - 1|$

- A $x \geq 4$
- B $x \geq -2$
- C $x \leq -2 \cup x \geq 4$
- D $-2 \leq x \leq 4$

Question 2

The polynomial $4x^3 + 4x^2 - 6x + 7$ has zeros α , β and γ .

What is the value of $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$

- A $-\frac{7}{4}$
- B $-\frac{4}{7}$
- C $\frac{4}{7}$
- D $\frac{7}{4}$

Question 3

Which of the following is true for $y = 3 - 4 \cos(\pi x + 3)$?

- A period = 2π amplitude = 4
- B period = 2 range = $-4 \leq y \leq 4$
- C range = $-1 \leq y \leq 7$ amplitude = 4
- D domain = $0 \leq x \leq 2\pi$ amplitude = 3

Question 4

In a drawer of socks there are 12 Red socks, 12 Blue socks and 4 Orange socks, all unmatched. Phoenix takes out individual socks, one at a time, in the dark. How many socks must Phoenix take out to be sure that he has at least two blue socks?

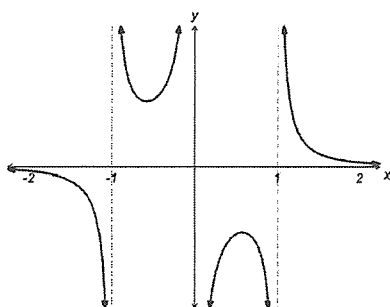
- A 4
- B 13
- C 17
- D 18

Question 5

The general solution to $2\cos 2x - 1 = 0$ is:

- A $x = n\pi \pm \frac{\pi}{6}$
- B $x = n\pi + (-1)^n \frac{\pi}{6}$
- C $x = 2n\pi \pm \frac{\pi}{3}$
- D $x = n\pi \pm \frac{2\pi}{3}$

Question 6



Above is the graph of the function $y = \frac{1}{f(x)}$

Which of the equations below best represents $f(x)$?

- A $f(x) = x^2 - 1$
- B $f(x) = x^3 - x$
- C $f(x) = x^4 - x^2$
- D $f(x) = x^2(x - 1)^2(x + 1)^2$

Section 11

60 marks

Attempt questions 7 – 12

Allow about 80 minutes for this section

Answer each question in your answer booklet. STARTING each NEW question at the TOP OF A NEW PAGE.

Your responses should include relevant mathematical reasoning and/ or calculations.

Question 7

10 marks

- a. Find the remainder when the polynomial $P(x) = 2x^4 - 3x^2 + 5x - 6$ is divided by $(x - 2)$. 1
- b. Differentiate the following with respect to x .
- i. $y = \frac{4}{\sqrt{x}}$ 2
- ii. $y = (x^3 - 1)^6$ 2
- c. Solve the inequality,
- $$2x^2 - 5x + 2 > 0$$
- 2
- d. Simplify, $\frac{1+\cot x}{1+\tan x}$ 2
- e. How many different arrangements of the letters from the word ASSESSMENT are possible if all 10 letters are used in each arrangement? 1

Question 8**10 marks**

Start this question at the TOP of a NEW page in your answer booklet.

- a. Find the equation of the tangent to the curve with parametric equation $x = 2t^3$ and $y = t^2 - t$ at the point where $t = 3$. 3

- b. Solve the inequality, 3

$$\frac{x-4}{x+2} \leq -1$$

- c. Solve, over the domain $0 \leq x \leq 2\pi$,

$$(\tan x - 1)(\sin x + 1) = 0$$
 2

- d. The line $3x + ay - 1 = 0$ and $bx - 2y + 9 = 0$ are perpendicular.
Find the value of $\frac{a}{b}$ 2

Question 9**10 marks**

Start this question at the TOP of a NEW page in your answer booklet.

- a. Find the number of ways that 7 students can be chosen from 6 senior students and 5 junior students,
- i. Without restrictions 1
 - ii. So that there are more senior students than junior students. 2
- b. Suppose that $x^3 + Ax^2 + 5x + B = (x + 3)(x - 2)Q(x) + 3$ for all x , where $Q(x)$ is a polynomial and A and B are real numbers.
- Find the values of A and B . 2
- c. Find the coefficient of x^{-4} in the expansion $\left(x^2 - \frac{3}{x}\right)^7$ 2
- d. A group of 7 friends, 4 girls and 3 boys are to be arranged in 7 seats.
- i. In how many ways can they sit around a circular table if the girls must sit together? 1
 - ii. What is the probability that the girls and boys take alternate seats if they are seated in a straight line? 2

Question 10**10 marks**

Start this question at the TOP of a NEW page in your answer booklet.

a. Express $\frac{d^2}{dx^2}(\sqrt{4+x^2})$ as a single simplified fraction. **3**

b.

i. Sketch the curve $g(x) = \frac{2x}{x+1}$, showing all intercepts and asymptotes.
Your sketch should be $\frac{1}{3}$ of a page. **2**

ii. Find the equation of $g^{-1}(x)$ and state the domain and range of the inverse function. **3**

iii. Find any values of x for which $g^{-1}(x) = g(x)$ **2**

Question 11**10 marks**

Start this question at the TOP of a NEW page in your answer booklet.

Consider the polynomial $P(x) = 4x^4 + 23x^3 + 42x^2 + 20x - 8$

The polynomial equation $P(x) = 0$ has a root of multiplicity 3.

- a. Find $P'(x)$ and $P''(x)$ and state the root of multiplicity 3. 2
- b.
- i. Express $P'(x)$ in the form,
 $(x + a)^2(bx + c)$ where a, b and c are integers 2
- ii. State the point on the curve $y = P(x)$ where the tangent to the curve is horizontal and $y \neq 0$. Leave your y -value for this point correct to the nearest integer. 1
- c. Express $P(x)$ as a product of its linear factors. 1
- d. Sketch the curve $y = P(x)$ showing all key features. 2
- e. On a separate set of axes, sketch the curve $h(x) = \sqrt{-P(x)}$,
Showing all key features. 2

Question 12

10 marks

Start this question at the TOP of a NEW page in your answer booklet.

- a. A bag contains k red marbles and one blue marble. Three marbles are drawn, without replacement, the probability of all three marbles being red is $\frac{5}{8}$.
How many red marbles are in the bag? 2

- b. A particle is moving in a straight line. The displacement, x metres, from the origin at time t seconds, is given by

$$x = \frac{t-1}{t+1}$$

- i. Find an expression for the velocity and acceleration of the particle at time t . 2
 - ii. Find the distance travelled by the particle in the first second of motion? 1
 - iii. Describe, by referencing displacement, velocity and acceleration, the motion of the particle as time elapses. 2
- c. Given that $(1+x)^5(1+x)^9 = (1+x)^{14}$, show that 3

$$\binom{9}{5} + \binom{5}{1}\binom{9}{4} + \binom{5}{2}\binom{9}{3} + \binom{5}{3}\binom{9}{2} + \binom{5}{4}\binom{9}{1} + \binom{9}{0} = \binom{14}{5}$$

End of task

Section 1

1 C

2 C

3 C

4 D

5 A

6 B

Section 1 ~ longer worked solns

$$3 \leq x - 1$$

$$2. a=4, b=4, c=-6, d=7$$

$$3 \leq x - 1 \text{ or } 3 \leq -x + 1$$

$$4 \leq x \quad x \leq -2$$

(C)

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{4} + \frac{1}{4} + \frac{1}{-6} = \frac{3}{12} + \frac{3}{12} - \frac{2}{12} = \frac{4}{12} = \frac{1}{3}$$

(C)

$$x \geq 4$$

$$= -5/a - d/a = \frac{-5}{4} - \frac{7}{4} = -\frac{12}{4} = -3$$

$$y = -4 \cos(\pi x + 3) + 3$$

$$\text{amp} = 4 \quad \text{period} = \frac{2\pi}{\pi} = 2$$

$$= 2$$

(C)

$$\text{range} = 4 + 3 \leq y \leq -4 + 3$$

$$-1 \leq y \leq 7$$

(D)

4. Pigeon hole = worse case

∴ all red, all orange then 2 Blue

$$= 12 + 4 + 2 = 18 \text{ socks}$$

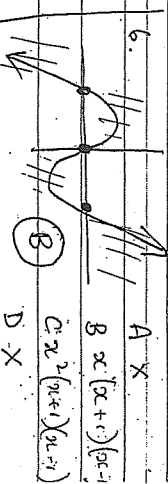
$$2 \cos 2x - 1 = 0$$

$$\cos 2x = \frac{1}{2}$$

$$2x = 2n\pi \pm \cos^{-1}(\frac{1}{2})$$

$$x = \frac{1}{2}(2n\pi \pm \pi/3)$$

$$x = n\pi \pm \pi/6 \quad (A)$$



Section 2

Question 7

$$a) P(x) = 2x^4 - 3x^2 + 5x - 6$$

$$P(2) = 2(2)^4 - 3(2)^2 + 5(2) - 6$$

$$= 24$$

$$b) \text{ remainder} = 24$$

$$i. y = 4x^{-1/2}$$

$$\frac{dy}{dx} = -2x^{-3/2}$$

$$= \frac{-2}{x\sqrt{x}}$$

$$ii. y = (x^3 - 1)^6$$

$$\frac{dy}{dx} = 6(x^3 - 1)^5 \cdot 3x^2$$

$$= 18x^2(x^3 - 1)^5$$

$$c) 2x^2 - 5x + 2 > 0$$

$$(2x - 1)(x - 2) > 0$$

$$x < 1/2, x > 2$$

$$d) \frac{1 + \cot x}{1 + \tan x}$$

$$= \frac{1 + \frac{1}{\tan x}}{1 + \tan x}$$

$$= \frac{\tan x + 1}{\tan x(1 + \tan x)}$$

$$= \frac{1}{\tan x}$$

$$= \cot x$$

e) ASSESSMENT 10 letters

$$4 \times S, A, M, N, T$$

$$10!$$

$$4! \times 2!$$

$$= 75600 \text{ arrangements}$$

Question 8

a. $x = 2t^3$ $y = t^2 - t$ $\rightarrow$ point $(2 \times 3^3, 3^2 - 3)$

$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$
 $= (2t - 1) \cdot \frac{1}{6t^2}$ at $t = 3$
 $= \frac{5}{54}$

$MT = 5, 1$ $\therefore 4 - 6 = 5$ $(x - 54)$
 $6(3)^2$ $\frac{54}{54}$

$= \frac{5}{54}$ $54y - 324 = 5x - 270$
 $5x - 54y + 54 = 0$

b. $\frac{x-4}{x+2} \leq -1$ $x \neq -2$

$(x-4)(x+2) \leq -1(x+2)^2$
 $(x-4)(x+2) + (x+2)^2 \leq 0$

$(x+2)[(x-4) + (x+2)] \leq 0$

$-(x+2)[2x-2] \leq 0$ $\rightarrow$ remember

$2(x+2)(x-1) \leq 0$

$\therefore -2 < x \leq 1$ or $[-2, 1]$

c. $(\tan x - 1)(\sin x + 1) = 0$

$\downarrow x = \frac{\pi}{4}, \frac{5\pi}{4}$ $x = \frac{3\pi}{2}$

But $x \neq \frac{3\pi}{2}$ as $\tan \frac{3\pi}{2}$ is undefined

$\therefore x = \frac{\pi}{4}, \frac{5\pi}{4}$ only

d. $ay = 1 - 3x$ $2y = 6x + 9$

$m_1 = -3/a$ $m_2 = b/2$

$\therefore -3/a \times b/2 = -1$ $a \perp b$

$b/a = 2/3$ $\therefore a = \frac{3}{2}$

Question 9

a. 6 senior 5 junior

${}^{11}C_7 = 330$

${}^{11}C_6 \times {}^5C_1 + {}^6C_5 \times {}^5C_2 + {}^6C_4 \times {}^5C_3 = 215$

b. $P(-3)$ and $P(2)$ have remainder 3

$\therefore -27 + 9A - 15 + B = 3$

$9A + B = 45$ --- (1)

and $8 + 4A + 10 + B = 3$

$4A + B = -15$ --- (2)

--- (1) - (2)

$5A = 60$

$A = 12$ $\therefore B = -63$

c. ${}^7C_k (x^2)^{7-k} (-3x^{-1})^k = x^{-4}$

equating

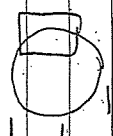
$14 - 2k + -k = -4$

$18 = 3k$

$k = 6$

$\therefore {}^7C_6 \times (-3)^6 = 5103$

d. $4 \times 3 \times 3$ $1 \times 4! \times 3! = 144$



$4! \times 3! = 144$

$\therefore$ Probability $= \frac{4! \times 3!}{7!} = \frac{144}{5040}$

Question 10

$$a. \frac{d}{dx} \left[\sqrt{4+x^2} \right] = \frac{1}{2} (4+x^2)^{-1/2} \cdot 2x$$

$$= \frac{x}{\sqrt{4+x^2}}$$

$$\frac{d^2}{dx^2} \left[\sqrt{4+x^2} \right] = \frac{\sqrt{4+x^2} (1) - (x) \cdot \frac{1}{2} (4+x^2)^{-1/2} \cdot 2x}{\sqrt{4+x^2}^2}$$

$$= \frac{\sqrt{4+x^2} - \frac{x^2}{\sqrt{4+x^2}}}{\sqrt{4+x^2}^2}$$

$$= \frac{4+x^2-x^2}{(4+x^2)^2 \sqrt{4+x^2}}$$

$$= \frac{4}{(4+x^2)^{5/2}}$$

$$a. g(x) = \frac{2x}{x+1} \quad \text{Inverse:}$$

$$ii) x = 2y$$

$$(y+1)x = 2y$$

$$xy + x = 2y$$

$$xy - 2y = -x$$

$$y(x-2) = -x$$

$$y = \frac{-x}{x-2}$$

$$\text{or } y = \frac{x}{2-x}$$

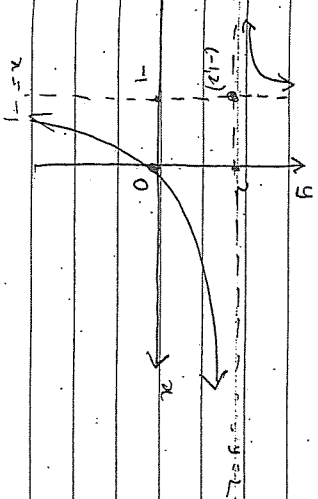
$$2: y \in \mathbb{R}, x \neq -1 \quad \text{or } y = \frac{x}{2-x}$$

$$1. \text{ take } f(x) = x \quad \rightarrow D: x \in \mathbb{R}, x \neq 2$$

$$\frac{2x}{x+1} = x \quad \rightarrow R: y \in \mathbb{R}, y \neq -1$$

$$2x = x^2 + x \quad \boxed{x=0} \quad \boxed{x=1} \text{ soln}$$

$$x^2 - x = 0 \quad x(x-1) = 0 \quad \therefore (0,0) \text{ or } (1,1)$$



Question 11

$$P(x) = 4x^4 + 23x^3 + 42x^2 + 20x - 8$$

$$a. P'(x) = 16x^3 + 69x^2 + 84x + 20$$

$$P''(x) = 48x^2 + 138x + 84$$

$$\text{Now } 48x^2 + 138x + 84 = 0$$

$$\text{test } 2x = -7/8 \text{ or } -2$$

$$P''(-2) = P'(-2) = P(-2) = 0$$

$$\therefore \text{root of multiplicity 3}$$

$$\text{is at } x = -2$$

$$b. 1. P'(x) = (x+2)^2 (16x+5)$$

$$\text{as } -2 + -2 + 5 = -69$$

$$x = -5/16$$

$$ii. (-5/16, -11) \text{ has a horizontal tangent}$$

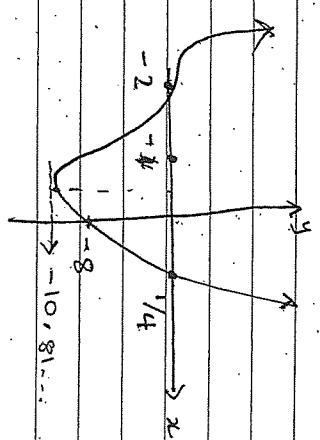
$$c. P(x) = (x+2)^3 (4x-1)$$

$$d. \quad e. y = \sqrt{-P(x)}$$

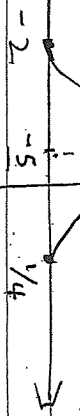
$$\text{invert } P(x)$$

$$\text{take the section only and } \sqrt{\quad}$$

$$\text{y values}$$



$$\text{Need T.P. and shape}$$



Question 12

a. k red 1 blue total = $k+1$

$$\therefore P(\text{red}) = \frac{k}{k+1} \cdot \frac{k-1}{k} \cdot \frac{k-2}{k-1} = \frac{5}{8}$$

$$\frac{k-2}{k+1} = \frac{5}{8}$$

$$8k-16 = 5k+5$$

$$3k = 21$$

$$k = 7$$

b. $x = t-1$

$t+1$

$$1. v = \frac{(t+1)(1) - (t-1)(1)}{(t+1)^2}$$

$$= \frac{2}{(t+1)^2}$$

$$a = -4(t+1)^{-3} \cdot 1$$

$$= \frac{-4}{(t+1)^3}$$

11. $t = 0$ $x = -1$

$t = 1$ $x = 0$ $\therefore$ distance 1 metre

111. as $t \rightarrow \infty$

$v \rightarrow 0^+$

motion of the particle

$x(a) \rightarrow 0^-$ approach limiting

$x \rightarrow +1$ position of 1

slowing down but not stopping.

Question 12

$$(c) (1+x)^5 (1+x)^9 = (1+x)^{14}$$

$$\text{RHS} = {}^{14}C_0 + {}^{14}C_1 x + {}^{14}C_2 x^2 + \dots + {}^{14}C_5 x^5 + \dots + {}^{14}C_{14} x^{14}$$

$$\text{LHS} = \left[{}^5C_0 + {}^5C_1 x + {}^5C_2 x^2 + \dots + {}^5C_5 x^5 \right] \left[{}^9C_0 + {}^9C_1 x + {}^9C_2 x^2 + \dots + {}^9C_9 x^9 \right]$$

co-efficient of x^5 on RHS is :

${}^{14}C_5$

co-efficient of x^5 on LHS is :

$${}^5C_0 \times {}^9C_5 + {}^5C_1 \times {}^9C_4 + {}^5C_2 \times {}^9C_3 + {}^5C_3 \times {}^9C_2 + {}^5C_4 \times {}^9C_1 + {}^5C_5 \times {}^9C_0$$

o.e. equating :

$${}^5C_0 {}^9C_5 + {}^5C_1 {}^9C_4 + {}^5C_2 {}^9C_3 + {}^5C_3 {}^9C_2 + {}^5C_4 {}^9C_1 + {}^5C_5 {}^9C_0 = {}^{14}C_5$$

as ${}^5C_0 = {}^5C_5 = 1$ then

$${}^9C_5 + {}^5C_1 {}^9C_4 + \dots + {}^9C_0 = {}^{14}C_5$$

Q.E.D.